

Custom stainless-steel vessel - 10 L default

custom-steel

Custom template

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	2 L	Custom template
Maximum working volume	10 L	Custom template
Nominal vessel / bag size	10 L	Custom template
Total vessel volume loaded into the form	10 L	Custom template
Initial working volume loaded into the form	2 L	Custom template
Product-contact material	Stainless steel	Custom template
Height : diameter	3	Custom template
Internal / chamber diameter	0.1619 m	Custom template
Liquid height at maximum working volume	0.4857 m	Custom template
Impeller / mixing type	Six-blade Rushton turbine	Custom template
Impeller count	3	Custom template
Impeller diameter	0.0534 m	Custom template
Power number	5	Custom template
Initial agitation	180 rpm	Custom template
Maximum agitation	500 rpm	Custom template
Initial aeration	0.5 vvm	Custom template
Maximum aeration	2 vvm	Custom template
Baffled vessel	Yes	Custom template
Maximum pressure	1 bar(g)	Custom template
Default operating pressure	0.2 bar(g)	Custom template
Maximum inlet oxygen fraction	100%	Custom template

Parameter evidence and derivations

Minimum working volume - 2 L

Custom template | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 2 L.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Maximum working volume - 10 L

Custom template | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 10 L.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Nominal vessel / bag size - 10 L

Custom template | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 10 L.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Total vessel volume loaded into the form - 10 L

Custom template | Form-population rule

Rule: total volume = max(nominal vessel volume, maximum working volume)

Selection / calculation: $\max(10 \text{ L}, 10 \text{ L}) = 10 \text{ L}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 2 L

Custom template | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(2 \text{ L}, 10 \text{ L}) = 2 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Stainless steel

Custom template | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Stainless steel.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 3

Custom template | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].heightDiameterRatio = 3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.1619 m

Custom template | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.01 \text{ m}^3) / (\pi \times 3)} = 0.161906 \text{ m}$; the form displays 0.1619 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.4857 m

Custom template | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.161906 \text{ m} \times 3 = 0.485718 \text{ m}$; the form displays 0.4857 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Six-blade Rushton turbine

Custom template | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].impellerType = "rushton"`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 3

Custom template | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].impellerCount = 3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.0534 m

Custom template | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.161906 \text{ m}, 0.001 \text{ m}) = 0.05342898 \text{ m}$; the form displays 0.0534 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 5

Custom template | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].powerNumber = 5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 180 rpm

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].baseRpm = 180

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 500 rpm

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].maxRpm = 500

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.5 vvm

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].baseVvm = 0.5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].maxVvm = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].baffled = true

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 1 bar(g)

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].maxPressure = 1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.2 bar(g)

Custom template | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].operatingPressure = 0.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

Custom template | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

All dimensions and operating limits should be replaced with the actual vessel data before simulation.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.